

Comments on Risk-Based Classification for Exercise 2

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1 Classification Based on Posterior Probabilities

The first part of the exercise deals with classification based on posterior probability. Given c classes, the probability that point $\mathbf{x}$ is in class ω_j is

$$P(\omega_j|\mathbf{x}) = \frac{p(\mathbf{x}|\omega_j)P(\omega_j)}{\sum_{i=1}^c p(\mathbf{x}|\omega_i)P(\omega_i)}. \quad (1)$$

The decision rule we use is

$$\arg \max_j (P(\omega_j|\mathbf{x})). \quad (2)$$

We can map the decision rule through a monotonically increasing function without changing the classification results. We do this to simplify our calculations. In this case, the $\ln(\cdot)$ function greatly simplifies things. We also drop the denominator term in (1) since it is a constant. Our resulting discriminant function is

$$g_j(\mathbf{x}) = \ln(p(\mathbf{x}|\omega_j)) + \ln(P(\omega_j)). \quad (3)$$

Because $\ln(\cdot)$ is a monotonically increasing function, if

$$P(\omega_1|\mathbf{x}) < P(\omega_2|\mathbf{x}) < \dots < P(\omega_c|\mathbf{x}) \quad (4)$$

then

$$g_1(\mathbf{x}) < g_2(\mathbf{x}) < \dots < g_c(\mathbf{x}). \quad (5)$$

Which means that a decision rule based on $g_j(\mathbf{x})$ is equivalent to one based on the posterior:

$$\arg \max_j (g_j(\mathbf{x})) \equiv \arg \max_j (P(\omega_j|\mathbf{x})). \quad (6)$$

2 Classification Based on Risk

Now we want to make a more flexible classifier which is based on the concept of risk. Suppose that we take an action α_i based on an observation $\mathbf{x}$. If the true state of nature is ω_j , then taking action α_i will have a cost of $\lambda(\alpha_i|\omega_j)$. The expected loss from action α_i , which is also called the conditional risk, is

$$R(\alpha_i|\mathbf{x}) = \sum_{j=1}^c \lambda(\alpha_i|\omega_j)P(\omega_j|\mathbf{x}). \quad (7)$$

If we select α_i so that $R(\alpha_i|\mathbf{x})$ is minimized, then our classifier will have optimal performance. For the purposes of this exercise, let's simplify things by constraining our possible actions to classifying an observation $\mathbf{x}$ to one, and only one, of c classes. Then

$$R_i(\mathbf{x}) = \sum_{j=1}^c \lambda_{ij}P(\omega_j|\mathbf{x}) \quad (8)$$

is the risk of classifying $\mathbf{x}$ to class i and λ_{ij} is the loss associated with selecting class i when $\mathbf{x}$ is actually in class j . Our decision rule is now

$$\arg \min_i (R_i(\mathbf{x})). \quad (9)$$

We can't simplify things in the same way as we did for posterior-based classification because

$$\ln(R_i(\mathbf{x})) = \ln \left(\sum_{j=1}^c \lambda_{ij}P(\omega_j|\mathbf{x}) \right) \neq \sum_{j=1}^c \ln(\lambda_{ij}P(\omega_j|\mathbf{x})). \quad (10)$$

(For example, $\ln(2+3) \approx 1.61 \neq \ln(2) + \ln(3) \approx 1.79$). We can however reuse (3) to create a risk-like discriminant function

$$r_i(\mathbf{x}) = \sum_{j=1}^c \lambda_{ij} \exp(g_j(\mathbf{x})) \quad (11)$$

which differs from (8) by a constant. This has the advantage of easy implementation since by this point, $g_j(\mathbf{x})$ is already coded. Risk-based classification can now be implemented according to

$$\arg \min_i (r_i(\mathbf{x})). \quad (12)$$

Note that $g_j(\mathbf{x})$ is in a logarithmic scale and $r_i(\mathbf{x})$ is in a linear scale. This means that a distance of ϵ used in the $g_j(\mathbf{x})$ should be mapped to $\exp(\epsilon)$ to preserve equivalency.

It may be tempting to create a discriminant function like

$$h_i(\mathbf{x}) = \sum_{j=1}^c \lambda_{ij} g_j(\mathbf{x}). \quad (13)$$

Unfortunately, this won't work. Both λ_{ij} and $g_j(\mathbf{x})$ may take on positive and negative values so (13) is not guaranteed to be monotonic with risk. I.e., if

$$R_1(\mathbf{x}) < R_2(\mathbf{x}) < \dots < R_c(\mathbf{x}) \quad (14)$$

then there is no guarantee that

$$h_1(\mathbf{x}) < h_2(\mathbf{x}) < \dots < h_c(\mathbf{x}) \quad (15)$$

is also true.